INCH-POUND

MIL-I-38391(USAF)
AMENDMENT 4
30 September 1991
SUPERSEDING
AMENDMENT 3
18 MARCH 1986

MILITARY SPECIFICATION

INDICATOR, ELECTRICAL TACHOMETER
ERU-12/A 0-350 RPM (ROTOR), 0-700
RPM (ENGINE)

This amendment forms a part of MIL-I-38391(USAF), dated 5 May 1965, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1 Add paragraph title: "Scope."

Delete 2.1 and add "2 1 Government documents."

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

* Under SPECIFICATION, Federal delete "DD-G-451", 'PPP-B-566", PPP-B-636", "PPP-B-665", "PPP-B-676", and add. "PPP-B-1672 Boxes, Shipping, Reusable With Cushioning "

* Under SPECIFICATION, Military, delete "MIL-P-7936", "MIL-L-25142", "MIL-M-26512', "MIL-R-26667', 'MIL-D-70327" entirely and add "DOD-D-100 Drawing, Engineering And Associated List' and MIL-B-131 Barrier Material, Water-Vaporproof, Flexible Heat Sealable '

* Under STANDARDS, Federal add. 'FED-STD-101 Preservation, Packaging and Packing Materials: Test Procedures."

* Under STANDARDS, Military delete "MIL-STD-143" entirely and add: 'MIL-STD-470 Maintainability Program Requirements (For Systems And Equipments)", "MIL-STD-471 Maintainability Demonstration", "MIL-STD-889 Dissimilar Metals", "MIL-STD-970 Standards & Specifications Order Of Preference For The Selection Of', and MIL-STD-2073/1A DOD Materiel Procedures For Development & Application Of Packaging Requirements."

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Add boxed paragraph at bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 by using the Standardization document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* Under STANDARDS, Military delete "MS33586 entirely.

* Delete the following parenthetical source statement and add new:

"(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)"

* Add: "2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AIR FORCE P/N 8988994
Luminescent material, Fluorescent"

Add. "2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society for Testing and Materials (ASTM)
ASTM C1036 Glass Flat, Standard Specification For 26 Apr 85

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add "2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detailed specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

* 3.2 Line 5 delete "MIL-STD-143" and add: "MIL-STD-970."

3.3.3 Line 2 delete "MS33586" and add: "MIL-STD-889."

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Add: "3.3.4.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item."

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3.4.3 Line 2 delete "MIL-M-26512" and add "MIL-STD-470

PAGE 4

Table I: In column entitled "SCALE ERROR TOLERANCE ± ROOM TEMPERATURE", delete "20" (three places) and substitue "30" and delete "10" (one place) and substitute "20".

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Table II: In column entitled "SCALE ERROR TOLERANCE ± ROOM TEMPERATURE", delete "20" (three places) and substitue "30" and delete "10" (one place) and substitute "20".

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Table III In column entitled "SCALE ERROR TOLERANCE ± ROOM TEMPERATURE opposite generator drive shaft speed 4100, delete "1" and substitute "2".

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Table IV In column entitled "SCALES ERROR TOLERANCE ± ROOM TEMPERATURE opposite generator drive shaft speed 4117, delete "1" and substitute "2".

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3.6 Line 6 delete "MIL-D-70327" and addd "DOD-D-100."

3.7 Line 4 delete "5.00 inches" and add "6.00 inches."

PAGE 9

* 3.8 Lines 2 and 3 delete "DD-G-451, type II quality AA" and adc "ASTM C1036, type I class I." Also line 5 delete "DD-G-451" and add: "ASTM C1036".

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Table V, delete "MIL-L-25142" and add: "Air Force P/N 8988994".

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Delete "4.1" entirely and add:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, contractor is responsible for the preformance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements."

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Add 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of maufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material."

Add "4.3.9 Order of tests. Unless otherwise specified, tests shall be conducted in the order shown under 4.6.

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4.5.2.2 Delete the test listing and add:

"a. Sampling plan A tests
b. Position error
c. Magnetic effect
d. Indicator current
e. Overspeed and reverse operation
f. Thermal shock
g. Vibration failure
h. Endurance
i. Salt fog

Note: See 4.3.8 for order of testing

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4.6.2 Add as second and third sentences: "The indicators shall be operated until their internal temperature has stabilized. For this test, the indicators shall be warmed up a minimum of 20 minutes. THIS WARMUP REQUIREMENT WILL NOT APPLY WHEN THE SCALE ERROR TEST IS APPLICABLE TO 4.6.7 AND 4.6.8."

4.6.5 Delete the second sentence and add: "Within 10 seconds after removal from the oven, the indicator cover glass shall be immersed to a depth of 0.25 inch in an ice-water bath maintained a 0° to 5°C." In the third sentence, delete "3 minutes" and add "5 minutes."

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4.6.6 Add as last sentence: "Evidence of fogging or damage shall be caused for rejection."

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4.6.20 Delete "MIL-R-26667" and add "MIL-STD-781."

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4.6.20.2 Delete and add. "4.6.20.2 Tests level The test level shall be in accordance with MIL-STD-781, test level F except that the indicators will be operated continuously at a generator drive shaft speed of 3000 - 4000 rpm and the low temperature portion of the temperature cycling shall be conducted at -35°C. Input voltages need not be monitored."

4.6.20.3.2 Delete "MIL-R-26667" and add. "MIL-STD-781."

4.6.21 Line 3 delete "MIL-M-26512" and add "MIL-STD-471."

Delete section 5 entirely and add:

* "5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each. Each item will be provided a preservation method IA-8 IAW MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073/1A to provide a method IA-8.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073/1A.

5.2 Packing. Packing shalll be level "A", "B", or "C" as specified (see 6.2).

5.2.1 Container-cushioning. Intermediate container requirements shall conform to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Intermediate containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate container. Cushioning material shall be of sufficient density and thickness to ensure shock transmission does not exceed peak values in G's established for the item.

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5.2 2 Level A. Item will be preserved as specified in 5.1 and shall be packe in exterior containers conforming to PPP-B-1672, weather resistant, unless otherwise specified by the contracitng activity, Exterior container shall be uniform shape, size, and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedures, as specified by contractor, special packaging instructions (SPI), and contracting activity.

5.2 3 Level B. Same as level "A' unless otherwise specified by special packaging instructions (SPI) or the contracting activity.

5.2.4 Level C. Item will be packed in containers in such a manner as to afford adequate protection against physical damage during direct domestic shipment from the supply sources to the first receiving activity. These packs shall conform to MIL-STD-2073/1A.

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5 3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, special packaging instructions (SPI), and MIL-STD-129.

5.4 Inspection and Test. Test of methods of preservation shall be accomplished in accordance with Section 4 of MIL-P-116 to ensure compliance with Section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.'

* Under section 6. NOTES add: '(This section contains information of a general o explanatory nature that may be helpful, but is not mandatory.)'

* Add' "6.2(f) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Preparing Activity.
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